



Analysis of atmospheric variations using GNSS signal as atmospheric sensor (PWV-GNSS) in the extreme rainfall events in Rio Grande do Sul (Brazil) in 2024

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Abstract

At the end of April and beginning of May 2024, the state of Rio Grande do Sul, Brazil, experienced an unprecedented climatic disaster. A combination of meteorological factors, including extreme accumulated precipitation, and the region's topography led to a rapid and severe rise in river levels. Therefore, numerous cities were inundated, leaving both people and animals homeless, and resulting in several fatalities. Given the magnitude of these events, this study aims to understand their underlying causes and explore strategies for identifying them both retrospectively and in near-real-time (nowcasting). To this end, we analyzed direct and indirect atmospheric measurements over the most affected areas in Rio Grande do Sul, particularly the cities of Porto Alegre and Santa Maria. Data were obtained from multiple sources, encompassing observations at the surface and throughout the atmospheric column. We mapped precipitable water vapor (PWV), precipitation, and flood extents. The recorded disasters were associated with accumulated rainfall levels that were over three times higher in Porto Alegre (457 mm) and five times higher in Santa Maria (719 mm) compared to previous El Niño events (2015 and 2016). Our findings underscore GNSS-derived PWV as a near-real-time predictor of extreme rainfall with up to 60 min of lead time.

Keywords Extreme weather events · GNSS meteorology · Precipitable water vapor (PWV) · Flood monitoring · Rio Grande do Sul disaster 2024

1 Introduction

The precipitation regime in the southern region of Brazil, particularly in the state of Rio Grande do Sul, is characterized by a relatively uniform distribution throughout the year. This pattern results from the influence of various meteorological systems that act across different seasons, continuously altering the region's weather conditions. These systems include

frontal systems, cyclones, Mesoscale Convective Complexes (MCCs), Upper Tropospheric Cyclonic Vortices (UTCVs), sea breeze circulations, and instability lines (Reboita et al. 2015).

In addition to these synoptic and mesoscale systems, interannual-scale phenomena also contribute to modulating the precipitation regime in the state of Rio Grande do Sul. One of the most prominent is the El Niño–Southern Oscillation (ENSO), a coupled ocean–atmosphere phenomenon in which changes in Sea Surface Temperature (SST) lead to variations in atmospheric pressure and wind patterns. These alterations influence large-scale atmospheric circulation and consequently affect the precipitation regime across several regions of the globe, including South America (Coelho et al. 2002; Marengo et al. 2012; Tedeschi and Collins 2016).

This phenomenon exhibits two distinct phases: El Niño, characterized by anomalous warming of the waters in the equatorial Pacific Ocean, and La Niña, marked by cooler-than-average SSTs in the same region. Both phases significantly influence the precipitation regime in the state of Rio Grande do Sul, with El Niño typically enhancing rainfall, while La Niña is often associated with drought conditions.

Between April and May 2024, the state of Rio Grande do Sul experienced unprecedented rainfall. According to the Brazilian National Institute of Meteorology (INMET), Santa Maria recorded an accumulated precipitation of 450.2 mm between April 29 and May 2, while Porto Alegre registered 297.2 mm during the same period. These values represent more than twice the climatological average expected for the entire month. The climatological normals for April and May in Santa Maria and Porto Alegre are 151.1 mm (114.4 mm) and 136.6 mm (112.8 mm), respectively, based on the 30-year average from 1991 to 2020. The 2024 rainfall event was classified as the most severe climate-related disaster in the state's history, marked not only by exceptionally high precipitation levels but also by widespread landslides and severe flooding.

According to Silveira et al. (2024), the northern region of Porto Alegre experienced the highest flood levels, reaching up to 6.0 m, particularly in the Jacuí River, one of the region's main waterways. These levels were exacerbated by the backwater effect caused by barriers. The observed water levels exceeded the design capacity of dikes constructed with an overflow threshold of 6.0 m and a safety margin of 1.25 m, which had been planned to follow one of the most significant floods in 1941 (maximum flood level of 4.75 m). Because of the 2024 events, numerous fatalities and disappearances were reported, alongside widespread displacement of people and animals, and the emergence of disease outbreaks (G1 2024; BBC News 2025; The New York Times 2024).

From a meteorological perspective, this extreme event was driven by a combination of factors acting at both climatic and synoptic scales. At the beginning of 2024, the region was under the influence of the El Niño phenomenon. During El Niño episodes, positive precipitation anomalies and more frequent storms are typically observed in southern Brazil (Cavalcanti et al. 2009; Araújo et al. 2023; Valente et al. 2023), along with an intensification of the Low-Level Jet (LLJ) (Silva and Ambrizzi 2006; Silva et al. 2009). The LLJ plays a key role in transporting moisture from the Amazon region to southern Brazil. Additionally, a cold front was present over the state of Rio Grande do Sul; however, its northward progression was blocked by an atmospheric blocking system over central Brazil, which prevented the front from advancing to lower latitudes.

Geodetic remote sensing satellites using Global Navigation Satellite Systems (GNSS) are a well-established technique for monitoring precipitable water vapor (PWV) (Gutman et al. 2004; Sapucci 2014; Sapucci et al. 2019). This approach exemplifies the synergy between Geodesy and Meteorology. Meteorology provides atmospheric profiles and forecasts of variable concentrations that are essential for modeling and minimizing the signal delay caused by atmospheric refractivity, thereby improving the accuracy of real-time GNSS positioning. Conversely, Geodesy, through GNSS technology, enables high temporal resolution (on the order of minutes) estimates of atmospheric parameters such as PWV (Gouveia et al. 2020). Moreover, the integration of GNSS, derived atmospheric products with Geographic Information Systems and Remote Sensing techniques has demonstrated significant potential to enhance spatial analyses and monitoring capabilities, particularly in applications related to climate variability and hydrometeorological hazards (Li et al. 2013; Huang et al. 2014).

In meteorology, the atmosphere is stratified based on temperature gradients into four primary layers, separated by three boundary surfaces. Meteorological phenomena occur primarily in the troposphere, the lowest layer, which extends from the Earth's surface up to approximately 18 km (Wallace and Hobbs 2006). In geodesy, GNSS satellite signals (radio waves) are affected as they propagate through the atmosphere to ground-based receivers. The Neutrosphere, which extends vertically from the surface to about 50 km altitude, is the atmospheric portion responsible for inducing errors, known as zenith total delay (ZTD). ZTD represents the cumulative delay experienced by GNSS signals as they propagate through the atmosphere, primarily due to the spatial and temporal variability of atmospheric pressure and moisture, which affect the refractive index along the signal path. This delay directly impacts positioning accuracy, causing an error of approximately 2.5 m in the zenith direction, and potentially up to ten times greater at low elevation angles (e.g., below 5°) (Mendes 1999; Nievinski and Santos 2010; Gouveia et al. 2020). The ZTD can be converted into PWV estimates, as detailed in the section PWV and GNSS Data.

In this way, we quantified the extreme rainfall and the variation of the Neutrosphere during the event, and we took measures that express this variation from different equipment and techniques. Among them are conventional meteorological stations of surface and high-altitude atmospheric soundings (radiosondes), remote sensing images, and GNSS data.

To quantify the extreme rainfall and the associated variations in the Neutrosphere during the event, we employed measurements derived from multiple instruments and methodologies. These included conventional surface meteorological stations, upper-atmosphere soundings (radiosondes), remote sensing imagery, and GNSS data.

This study aims to analyze the response of neutral atmospheric variations and zenithal delay (ZTD) during the extreme rainfall events that occurred in Rio Grande do Sul (Brazil) in April and May 2024. The results and analyses based on different data sources are presented in Sects. 3 to 7. Section 3 addresses the variability of atmospheric parameters using a 10-year time series; Sect. 4 presents the ZTD derived from radiosonde data; and Sect. 5 focuses on remote sensing observations. Finally, Sects. 6 and 7 provide a discussion of the results and the final considerations, respectively.

2 Data and methodology

The dataset used in this study integrates multiple data sources, reflecting both the complexity of modeling variations in the neutral atmosphere, particularly during extreme precipitation events, and the limitations imposed by data gaps resulting from the impacts of the disaster. The primary analysis focused on the period from April 6 to May 6, 2024, encompassing conditions before and after the extreme precipitation event that occurred between April 26 and May 5. For the climatological context, a 10-year time series (2015–2024) was considered for the same temporal window (April 6 to May 6).

The integration of multiple data sources enabled the analysis of the Neutrosphere across different altitudinal layers and ensured continuity in the dataset when information was missing from any individual source. The data and methodology are organized into four sections: (i) study areas; (ii) surface meteorological stations (INMET 2024) and upper-air observations from radiosondes (with measurements from the surface up to approximately 28 km) (University of Wyoming 2023), as detailed in the Meteorological Data section; (iii) PWV/GNSS data, which provide ZTD estimates from the top of the Neutrosphere (~50 km) to the surface; and (iv) optical imagery from remote sensing satellites, with observations captured at an altitude of approximately 790 km.

2.1 Study areas

In Fig. 1, the selected study areas are presented, covering the two cities most affected by the tragedy: Brazil (a), in the state of Rio Grande do Sul (b) Porto Alegre and Santa Maria (c). The average altitudes of the cities of Porto Alegre and Santa Maria (Fig. 1(c)) are 10 m and 115 m above sea level, respectively.

Figure 1 presents the selected study areas, highlighting the two cities most severely affected by the disaster: Brazil (Fig. 1a), in the state of Rio Grande do Sul (Fig. 1b), Porto Alegre and Santa Maria (Fig. 1c) municipalities. The mean altitudes of Porto Alegre and Santa Maria are approximately 10 m and 115 m above sea level, respectively. The elevation data shown in Fig. 1 were obtained from the ANADEM (National Water Agency Digital Elevation Model) (Laipelt et al. 2024). ANADEM is a corrected 30-m resolution digital terrain model (DTM) developed by Federal University of Rio Grande do Sul (UFRGS) and Brazilian National Water Agency (ANA), designed to reduce bias from vegetation cover. According to Laipelt et al. (2024), ANADEM achieves a vertical root mean square error (RMSE) of approximately ± 5 m when validated against field survey control points across varying Brazilian terrain types. The data can be accessed at <http://ufrgs.br/hge/anadem>.

2.2 Surface and altitude meteorological data

Meteorological data were obtained from automatic surface stations and radiosondes. Surface measurements were sourced from the Brazilian National Institute of Meteorology (INMET 2024), specifically from the automatic stations located in Porto Alegre (A801) and Santa Maria (A803). Hourly data on temperature, relative humidity, and precipitation were analyzed for the period from April 1 to May 6 over the past 10 years. Additionally, precipitation data from more than 43 INMET stations across Rio Grande do Sul were utilized for the months of April and May 2024. The identification codes and geographic coordinates of

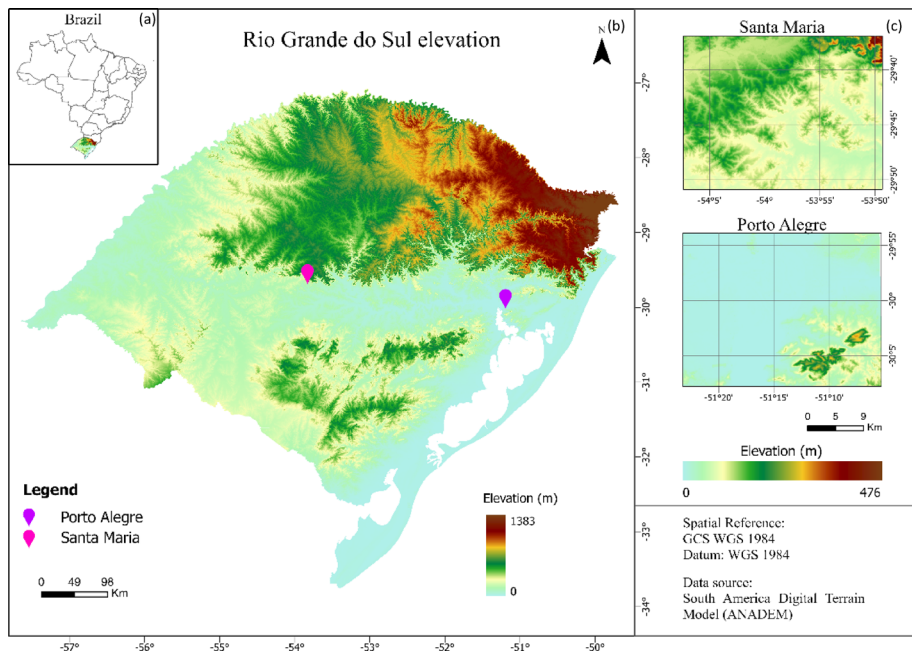


Fig. 1 Location of the study areas—Brazil (a), the state of Rio Grande do Sul (b), Santa Maria and Porto Alegre municipalities (c)—showing the state's elevation, and the respective study areas according to ANADEM data. MaySource: Author (2024)

the 45 stations used in the analysis are summarized in Table 1 and spatially represented in Fig. 3.

The radiosonde data were obtained from University of Wyoming (University of Wyoming 2023) (<http://www.weather.uwyo.edu/upperair/sounding.html>). A radiosonde is a meteorological instrument carried by a balloon to collect atmospheric data at different altitudes. Radiosonde profiles collect measurements from the balloon's launch point at the Earth's surface up to its burst altitude, approximately 28 km. During its ascent, it records variables such as temperature (T, in °C), atmospheric pressure (P, in hPa), relative humidity (RH—in %), and mixing ratio (MIXR—in g/kg) providing a detailed vertical profile of atmospheric conditions. These high-resolution observations are essential for weather forecasting, climate monitoring, and atmospheric research (Durre et al. 2018). In this study, three of these variables, temperature, pressure, and relative humidity, were utilized for further analysis.

Although radiosondes are highly effective tools for investigating extreme weather events, their deployment can be hindered under adverse conditions. During the extreme events in Rio Grande do Sul, particularly in Porto Alegre and Santa Maria, intense rainfall and widespread flooding significantly disrupted radiosonde operations. As a result, atmospheric profiling and the derivation of products such as radiosonde-based PWV were limited due to a substantial number of missed launches and data gaps.

Table 2 presents the geometric (Geo_Dist, in km) and altimetric (Alt_Dist, in m) distances between the meteorological stations in Porto Alegre and Santa Maria. These values enabled the analysis of atmospheric variable variations based on surface data collected at different altitudes. Notably, in Porto Alegre, the maximum geometric distance between the

Table 1 INMET station codes and coordinates (latitude, longitude, altitude)

City	Code	Latitude	Longitude	Altitude
Santa Vitória Do Palmar	A899	−33.74	−53.37	7.41
Jaguarão	A836	−32.53	−53.38	31.48
Rio Grande	A802	−32.08	−52.17	4.92
Capão Do Leão (Pelotas)	A887	−31.80	−52.41	13.00
Cangucu	A811	−31.40	−52.70	446.81
Bagé	A827	−31.35	−54.01	226.19
Mostardas	A878	−31.25	−50.91	3.82
Dom Pedrito	A881	−31.00	−54.62	150.00
Camaquã	A838	−30.81	−51.83	92.30
Santana Do Livramento	A804	−30.75	−55.40	196.00
Cacapava Do Sul	A812	−30.55	−53.47	420.82
Encruzilhada Do Sul	A893	−30.54	−52.52	427.75
Quaraí	A831	−30.37	−56.44	113.05
São Gabriel	A832	−30.34	−54.31	114.89
Porto Alegre-Belém Novo	B807	−30.19	−51.18	3.30
Porto Alegre—Jardim Botânico	A801	−30.05	−51.17	41.18
Tramandaí	A834	−30.01	−50.14	4.56
Rio Pardo	A813	−29.87	−52.38	106.99
Uruguaiana	A809	−29.84	−57.08	74.29
Santa Maria	A803	−29.72	−53.72	103.10
Alegrete	A826	−29.71	−55.53	120.88
São Vicente Do Sul	A889	−29.70	−54.69	134.00
Campo Bom	A884	−29.67	−51.06	23.35
Teutônia	A882	−29.45	−51.82	81.00
Canela	A879	−29.37	−50.83	830.93
Torres	A808	−29.35	−49.73	8.44
Santiago	A833	−29.19	−54.89	390.03
Bento Gonçalves	A840	−29.16	−51.53	623.27
Tupanciretã	A886	−29.09	−53.83	462.00
Cambará do Sul	A897	−29.05	−50.15	1017.00
Soledade	A837	−28.86	−52.54	660.44
São José Dos Ausentes	A829	−28.75	−50.06	1228.59
Serafina Corrêa	A894	−28.70	−51.87	485.29
Ibirubá	A883	−28.65	−53.11	455.27
São Borja	A830	−28.65	−56.02	81.08
Cruz Alta	A853	−28.60	−53.67	426.69
Vacaria	A880	−28.51	−50.88	969.89
São Luiz Gonzaga	A852	−28.42	−54.96	245.50
Passo Fundo	A839	−28.23	−52.40	680.67
Lagoa Vermelha	A844	−28.22	−51.51	833.83
Palmeira Das Missões	A856	−27.92	−53.32	614.11
Santa Rosa	A810	−27.89	−54.48	272.84
Santo Augusto	A805	−27.85	−53.79	489.67
Erechim	A828	−27.66	−52.31	777.08
Frederico Westphalen	A854	−27.40	−53.43	489.42

Table 2 Geometric distance (km) (Geo_Dist) and absolute altitude difference (m) (Alt_Dist) between surface meteorological stations (INMET) and altitude stations (radiosondes) in Porto Alegre and Santa Maria and their identification codes (Code)

City	INMET station	Radiosondes station	Geo_Dist (km)	Alt_ Dist (m)
	Code	Code		
Santa Maria	A803	SBSM	2.06	18.10
Porto Alegre	A801	SBPA	5.98	38.18

INMET and radiosonde stations was 6 km, with an altimetric difference of approximately 38 m.

2.3 PWV and GNSS data

The ZTD estimated from GNSS (ZTD/GNSS) quantifies Neutrospheric delay experienced by signals received at a GNSS station with known geographic coordinates. This technique offers high temporal resolution, ranging from one to five minutes, and continuous temporal coverage (24 h a day, 365 days a year). Although GNSS satellites orbit at altitudes exceeding 20,000 km above the Earth's surface. The ZTD is predominantly induced by the signal's propagation through the Neutrosphere, which extends up to approximately 50 km. For further details on ZTD/GNSS estimation, refer to Nievinski and Santos (2010), Elgered and Wickert (2017), and Gouveia et al. (2020).

There are networks of regional and local GNSS stations worldwide that provide this data free of charge. The Santa Maria and Porto Alegre stations belong to the Brazilian Network for Continuous Monitoring of GNSS Systems (RBMC) (RBMC, 2024).

The IWV can be obtained from the integral absolute humidity (ρ_w) along the vertical column from the surface (h_0) up to the altitude (h) as expressed by Eq. (1) (Bevis et al. 1992; Elgered and Wickert 2017; Gouveia et al. 2020):

$$IWV = \int_{h_0}^h \rho_w dh = d_{nh}^z \cdot \Psi. \quad (1)$$

Determining this integral requires the use of sophisticated techniques and equipment. So, using the ZTD known from GNSS estimation (Mendes 1999; Nievinski and Santos 2010; Gouveia et al. 2020) it is possible to obtain the PWV/GNSS (second part of Eq. 1), from the relationship given by Ψ (kg/m³):

$$\Psi = \frac{10^6}{R_w \left[k'_2 + \frac{k_3}{T_m} \right]}, \quad (2)$$

where R_w is the molar mass (kg) of the water molecule and the Neutrosphere average temperature (T_m) (K) (Davis et al. 1985; Bevis et al. 1992; Sapucci 2014).

PWV represents the total amount of water vapor contained within a vertical column of the atmosphere, expressed in millimeters of equivalent liquid water. This parameter provides a physically intuitive measure of atmospheric moisture content and can be directly compared with precipitation estimates derived from other observational sources, such as those discussed in previous sections. PWV (determined in millimeters) can be determined from IWV by applying a scale factor, the density of liquid water (ρ_a) (1 kg/m³) (Eq. 3). If

the IWV is determined from the IWV/GNSS, the PWV from Eq. 3 is called the PWV/GNSS (Bevis et al. 1992; Elgered and Wickert 2017; Gouveia et al. 2020):

$$PWV = \frac{IWV}{\rho_a} \quad (3)$$

The IWV and consequently the PWV estimated from GNSS measurements in relative positioning were determined from the Bernese scientific software (Bern 2015). The main files used to obtain the ZTD in Bernese are observation data, mapping function data (VMF1), precise ephemeris files, Earth rotation parameters, ionosphere, and coordinate files (Bern 2015).

Table 3 shows the horizontal and vertical distances between the RBMC stations selected in the cities of Santa Maria (SMAR) and Porto Alegre (POAL) and the nearest INMET.

The PWV obtained from radiosonde data is considered a reference for evaluating PWV-GNSS due to its ability to collect in situ atmospheric information. Radiosonde data will be processed using the automatic Neutrosphere Processing Tool (NEPTool), to calculate the zenith delay and PWV (De Lima et al. 2022; Albuquerque et al. 2024; Gouveia et al. 2025).

Precipitable Water Vapor (PWV) represents the total amount of water vapor contained in a vertical column of the atmosphere if condensed into liquid form. Physically, PWV expresses the atmosphere's capacity to generate precipitation, being directly related to moisture convergence, evapotranspiration, and horizontal vapor transport by large-scale circulation (Shao et al. 2023). A rapid increase in PWV indicates an intensification of moisture availability in the lower troposphere, promoting deep convection and the development of heavy precipitation systems (Zhang et al. 2025). From an operational perspective, continuous monitoring of PWV, particularly through high-temporal-resolution GNSS observations, enables the early identification of pre-convective environments and supports nowcasting applications. This integration of PWV data into forecasting systems enhances short-term prediction capabilities and improves understanding of the coupling processes between the surface and the atmosphere (Sapucci et al. 2019).

2.4 Optical images from remote sensing satellites

The optical imagery used in this study was acquired from the Sentinel-2 mission, part of the European Space Agency's (ESA) Copernicus Program. This mission comprises a constellation of two satellites, Sentinel-2A (S2A) and Sentinel-2B (S2B), operating in tandem and positioned 180° apart in a Sun-synchronous orbit at an average altitude of 786 km. Together, they provide global coverage with a revisit frequency of five days.

Sentinel-2 imagery is captured by the MultiSpectral Instrument (MSI), which offers 13 spectral bands at varying spatial resolutions. The data is openly and freely accessible, sup-

Table 3 Geometric distance (km) (Geo_Dist) and absolute altitude difference (m) (Alt_Dist) between INMET stations and GNSS in Porto Alegre and Santa Maria and their identification codes (Code)

City	INMET station Code	GNSS station Code	Geo_Dist (km)	Alt_ Dist (m)
Santa Maria	A803	SMAR	5.75	0.05
Porto Alegre	A801	POAL	0.77	1.47

porting a wide range of institutional and research applications. Further technical details on the sensor's specifications are available in ESA (2024a).

The data were acquired as close as possible to the April–May 2024 flood event, ensuring cloud-free conditions and minimal influence from prior flooding. This approach aimed to accurately capture the original extent of water bodies and their spatial definitions.

To calculate the water-covered area, representing the original delineation of water bodies, data from MapBiomas Collection 8 (2022) were utilized (Mapbiomas 2024). The MapBiomas Project, an initiative of the Climate Observatory, is developed through a collaborative network involving universities, Non-Governmental Organizations (NGOs), and technology companies. Its primary objective is to produce annual maps of land cover and land use in Brazil and to monitor changes in the national territory over time.

To determine the extent of areas covered by water following the intense rainfall period, Sentinel-2 Level-2A images with bottom-of-atmosphere reflectance values were acquired for both Santa Maria and Porto Alegre, dated May 6, 2024. The Modified Normalized Difference Water Index (MNDWI), as proposed by Xu (2006), was computed using atmospherically corrected bands Green (Band 3, 10 m) and Short-Wave InfraRed (SWIR) (Band 11, 20 m), which were resampled to a common resolution of 10 m. A thresholding technique was then applied to distinguish water bodies from other land cover types. The MNDWI is calculated according to Eq. 4. The resampling and MNDWI calculation were performed using Sentinel Application Platform (SNAP) 9.0.

$$MNDWI = \frac{\rho_{Green} - \rho_{SWIR}}{\rho_{Green} + \rho_{SWIR}} \quad (4)$$

where ρ_{Green} and ρ_{SWIR} are the reflectance of the Green and SWIR bands, respectively.

MNDWI allows highlighting water features and minimizing the rest of the targets, considering that the green band is located in the region of high water reflectance and the SWIR band in the region of high absorption of water contents.

Many previous research works have demonstrated that MNDWI is a suitable spectral index to enhance water information and extract water bodies, such as the identification of watercourses (Li et al. 2013; Du et al. 2014), mapping of flooded areas (Huang et al. 2014; Nandi et al. 2017), among other applications. The value resulting from the MNDWI calculation (Eq. 4) varies from -1 to 1. To extract water features from MNDWI images, a threshold of $MNDWI \geq 0$ was applied. This thresholding criterion follows the standard proposed by Xu (2006), who demonstrated that this value effectively distinguishes open water surfaces from built-up and vegetated areas in atmospherically corrected imagery. The method has since been widely adopted in the literature (Rokni et al. 2014; Feyisa et al. 2014) due to its simplicity, generalizability, and consistently high performance across diverse land cover types and climatic contexts.

Sentinel-3 is a European Earth Observation satellite mission developed to support a wide range of applications within the Copernicus program, including ocean, land, atmospheric, emergency, security, and cryosphere monitoring. The mission is operated jointly by the European Space Agency (ESA) and the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT).

Besides Sentinel-2, Sentinel-3 is also a satellite developed by ESA under the Copernicus program, designed for environmental and oceanographic monitoring. The Sentinel-3 mis-

sion comprises two satellites, Sentinel-3A and Sentinel-3B, in a near-polar, sun-synchronous orbit with an inclination of 98.65° .

The two operational satellites are equipped with the Ocean and Land Colour Instrument (OLCI), a multispectral sensor operating in 21 spectral bands, providing a revisit time of less than two days at the equator with a spatial resolution of 300 m across all surfaces.

Among the products, the instrument provides atmospheric by-products such as Integrated Water Vapour (IWV) columns, which represent the total amount of water vapor integrated over a column of the atmosphere, expressed in kg/m^2 . This measurement uses the input bands Oa18 (885 nm) and Oa19 (900 nm) (ESA 2024b).

For this study, an IWV OLCI Level-2 Land Full Resolution (LFR) image, taken on May 9, 2024, was acquired to represent the spatial distribution of IWV over the municipalities of Santa Maria and Porto Alegre.

The diverse datasets analyzed in this study enabled a comprehensive assessment of the Neutrosphere from multiple perspectives. In situ measurements of atmospheric variables such as temperature, relative humidity, and precipitation were obtained from automatic surface meteorological stations. Atmospheric profiles from radiosonde launches provided vertical observations from the surface up to approximately 28 km. GNSS data offered estimates of the ZTD, which were subsequently converted into PWV, representing the integrated water vapor content throughout the Neutrosphere down to the station altitude. Additionally, optical remote sensing imagery contributed spatial insights into water vapor distribution from satellite observations.

3 Results and analysis

The results are organized into three main sections. The first section presents atmospheric variability based on climatological data collected from April 1 to May 6 over the period 2015–2024. The second section analyzes PWV estimates derived from GNSS and radiosonde data during the critical event window, from April 27 to May 7, 2024. Finally, the third section examines the extent of surface water coverage in the study areas and the IWV retrieved from optical satellite imagery acquired during the flood event.

3.1 Atmospheric variation

To analyze atmospheric variation and the impact of extreme rainfall events, Fig. 2 presents the average of temperature (a), relative humidity (b), and precipitation (c) for the period from April 1 to May 6 from 2015 to 2024 for the stations located in Porto Alegre (purple bar) and Santa Maria (green bar). The light colors indicate El Niño (blue) and La Niña (pink) events.

Between 2015 and 2024, the average temperature in Porto Alegre was 20.9°C , while in Santa Maria it was 19.6°C . The maximum temperature recorded was 23°C in Porto Alegre in 2018, one degree Celsius higher than that observed in Santa Maria during the same year. The temperature range over the period was 3.46°C in Porto Alegre and 3.97°C in Santa Maria. The difference in average temperatures between El Niño and La Niña periods was minimal, not exceeding 0.15°C at either station. However, the observed patterns of relative

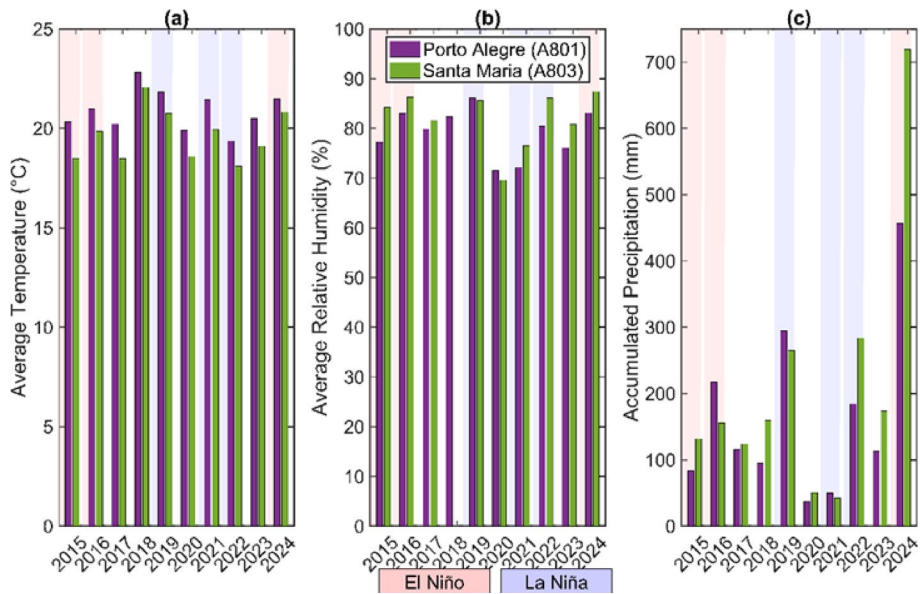


Fig. 2 Average of temperature (a) in °C, relative humidity (b) in %, and precipitation (c) in mm for Porto Alegre (purple bar) and Santa Maria (green bar) in the last 10 years in the period from April 1 to May 6. Highlighted periods of El Niño (blue) and La Niña (pink) events. MaySource: Author (2024)

humidity and precipitation did not fully correspond to the expected climatological behavior typically associated with these phases of ENSO.

Relative humidity values were consistently high across both regions, with Santa Maria reaching a maximum of 87.38% in 2024 and a minimum of 69.56% in 2020. Notably, precipitation anomalies in 2024 were substantial. In Porto Alegre, total rainfall reached 457 mm, representing a +203.86% increase compared to the average during previous El Niño years (2015 and 2016), and a +245.69% increase relative to the average of the past nine years (2015–2023). In Santa Maria, precipitation reached 719.6 mm—approximately five times the average observed during earlier El Niño events and exceeding the average of the previous 9 years by +367.48%. While elevated temperature and humidity are expected for southern Brazil, all meteorological parameters reached their peak values in 2024, with precipitation showing the most significant deviation.

Figure 3 presents the accumulated precipitation recorded by 45 automatic meteorological stations of the INMET network (INMET 2024) across the state of Rio Grande do Sul during April and May 2024. The gray bars represent the total precipitation observed in 2024, while blue and red bars indicate positive and negative anomaly percentages, respectively, relative to the climatological average. These anomalies were computed by comparing 2024 observed values with the 30-year climatological average (1991–2020), derived from 18 nearby conventional INMET stations. In red, precipitation is below average.

This comparison enables the evaluation of rainfall anomalies by contrasting the recorded values with the region's expected climatological behavior. It is important to highlight that some stations did not report precipitation measurements due to data unavailability during the studied period (e.g., station A878 in May), and the limited spatial resolution of conven-

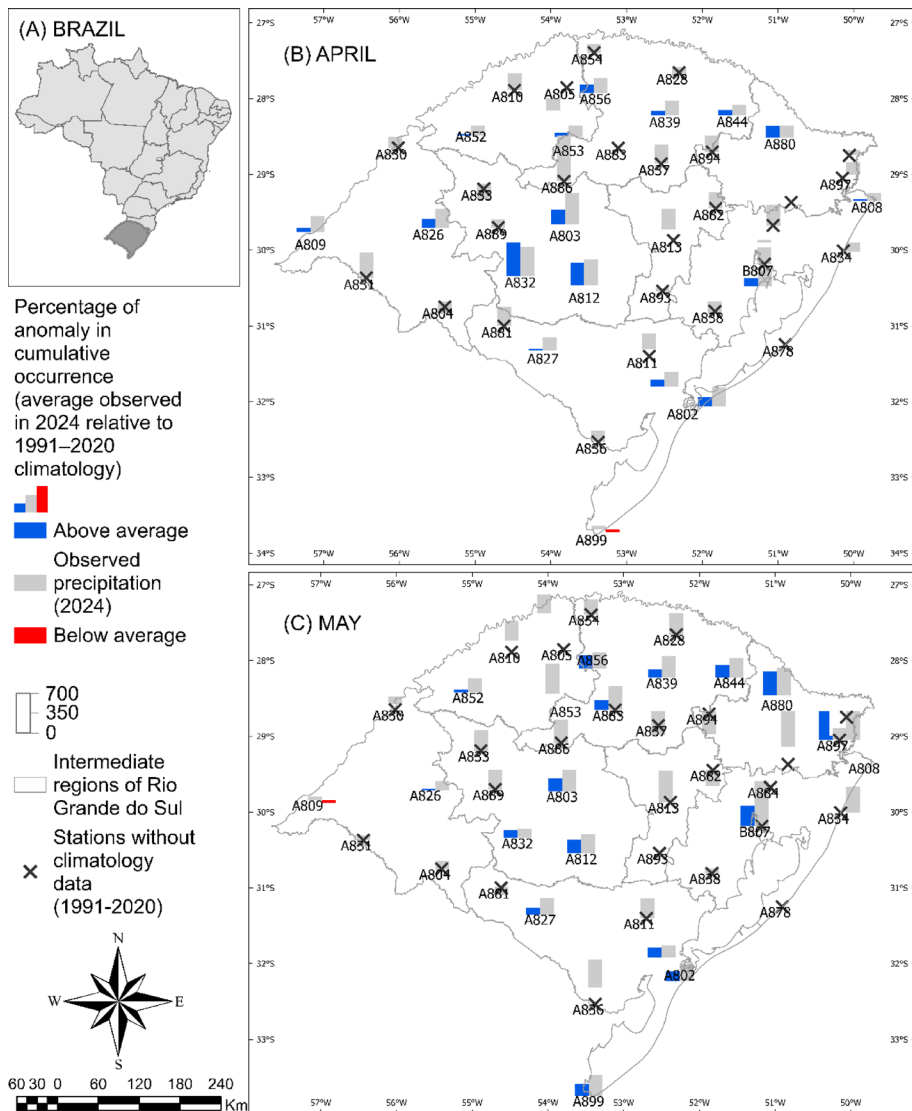


Fig. 3 Map of accumulated precipitation from INMET's automatic stations in the intermediate regions of Rio Grande do Sul, for the months of April and May 2024, and their respective anomaly percentages. MaySource: Author (2024)

tional stations prevented the calculation of anomalies for several locations. These stations are marked with an “X” on the map.

A general analysis of accumulated precipitation from INMET stations (Fig. 3) reveals that rainfall in May 2024 was predominantly concentrated in the northern and central regions of Rio Grande do Sul, whereas April exhibited a more widespread spatial distribution. In terms of total volume, May registered a higher accumulated precipitation (16.140 mm) compared to April (12.189 mm), with an excess of 3.951 mm. When compared to the climatological

normals, precipitation in both April and May 2024 significantly exceeded historical averages across nearly all monitored locations. Exceptions were observed in Santa Vitória do Palmar (automatic station A899 and conventional station 83,997) in April, as well as in Uruguiana during May, where precipitation levels remained below or within the climatological expectations.

Furthermore, Santa Maria (automatic station A803 and conventional station 83,936) recorded the highest accumulated rainfall in April 2024 (520.4 mm), exceeding climatology (369.3 mm) by +244.41%. In May 2024, the city with the most rainfall was Soledade (737.8 mm—A837), followed by Serafina Corrêa (714.4 mm—A894), Canela (706.0 mm—A879), Bento Gonçalves (688.6 mm—A840), Cambara do Sul (618.2 mm—A897), Rio Pardo (561.4 mm—A813), Ibiruba (524.6 mm—A883) and Porto Alegre (524.4 mm—A801).

3.2 PWV from direct measurements of the radiosonde and GNSS estimation

Figure 4 shows the comparison of PWV derived from GNSS observations and radiosonde (RDS) data for the stations of Porto Alegre and Santa Maria. In both locations, a strong linear correlation is observed between the two datasets, with correlation coefficients (R) exceeding 0.9, indicating excellent agreement. The bias is 2.37 mm in Porto Alegre and 0.72 mm in Santa Maria, suggesting that GNSS slightly underestimates PWV at Porto Alegre, while the estimates in Santa Maria are nearly unbiased. The root mean square error (RMSE) and standard deviation (STD) indicates low dispersion in both locations, with RMSE = 2.40 mm and STD = 3.45 mm in Porto Alegre, and RMSE = 2.70 mm and STD = 2.68 mm in Santa Maria. These results highlight the high consistency and reliability of GNSS-derived PWV compared with radiosonde observations.

PWV data are inherently time series and therefore exhibit strong temporal autocorrelation. To visually compare the distributions of PWV derived from GNSS and RDS measurements, boxplots were employed. This approach enables a direct evaluation of differences

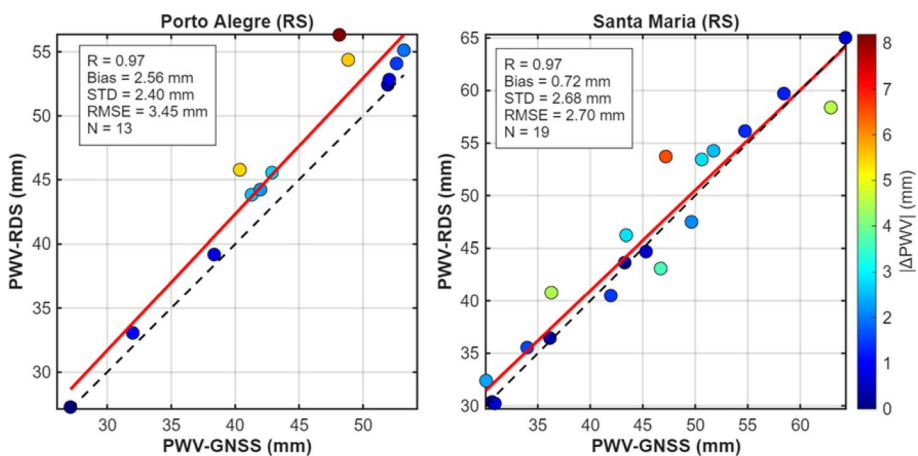


Fig. 4 Statistical validation showing correlation coefficients (R), bias, RMSE, and STD between PWV derived from GNSS (PWV-GNSS) and radiosonde (PWV-RDS) data at the stations of Porto Alegre and Santa Maria. MaySource: Author (2025)

in central tendency (median) and variability (dispersion) between both measurement techniques, while also allowing clear identification of potential outliers.

Figure 5 presents boxplots comparing PWV-GNSS and PWV-RDS distributions for Porto Alegre and Santa Maria. In Porto Alegre, the analysis was restricted to 13 RDS data points because the airport station was closed on May 3 due to flooding. Although the interquartile ranges (IQRs) differ [GNSS: (24.48, 58.96); RDS: (33.06, 56.35)], the medians remain nearly identical, as shown in the figure. A notable outlier is observed in the RDS dataset on April 28 (27.25 mm); however, the corresponding GNSS value (27.13 mm) suggests that this point is a statistical artifact caused by limited sampling. In Santa Maria, where 19 RDS data points were available, the IQRs display greater similarity [GNSS: (27.60, 65.92); RDS: (30.24, 65.01)], and a strong agreement between the medians is maintained.

Since GNSS instrumental errors are not easily quantifiable, and aggregating hourly data into daily averages would obscure short-term variations, a metric of local temporal variability was adopted. The bars in the GNSS plot represent a moving standard deviation computed over a 6-h sliding window. These bars do not indicate measurement uncertainty but rather depict signal instability within the corresponding time period. Larger bars denote intervals of rapid atmospheric change or high PWV variability, whereas smaller bars reflect more stable conditions. For RDS data, due to the limited number of observations, the visualization of this approach was not effective; therefore, only the individual data points were retained in the plot.

Figure 6 displays the PWV estimated from radiosonde data, alongside PWV-GNSS values derived from the ZWD obtained through GNSS data processing using the Bernese GNSS Software, version 5.2. The conversion of ZWD to PWV was carried out using Eqs. (1), (2), and (3), as detailed in the methodology section. Additionally, Fig. 6 includes the accumulated precipitation recorded by INMET automatic weather stations located in the municipalities of Porto Alegre and Santa Maria, in the state of Rio Grande do Sul, during the period from April 27 to May 7, 2024.

The PWV data reveals a clear relationship with the precipitation events recorded in both cities. A strong temporal association was observed between rising PWV and subse-

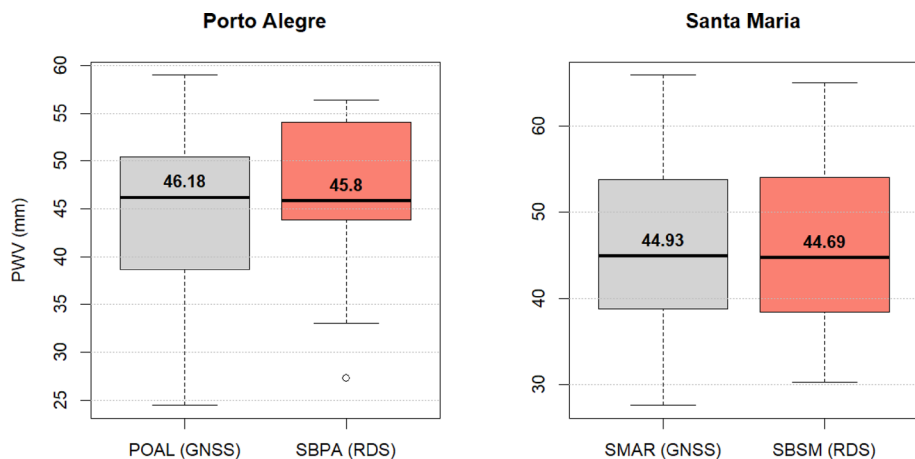


Fig. 5 Boxplot to compare the distributions of PWV-GNSS and PWV-RDS to Porto Alegre and Santa Maria. MaySource: Author (2025)

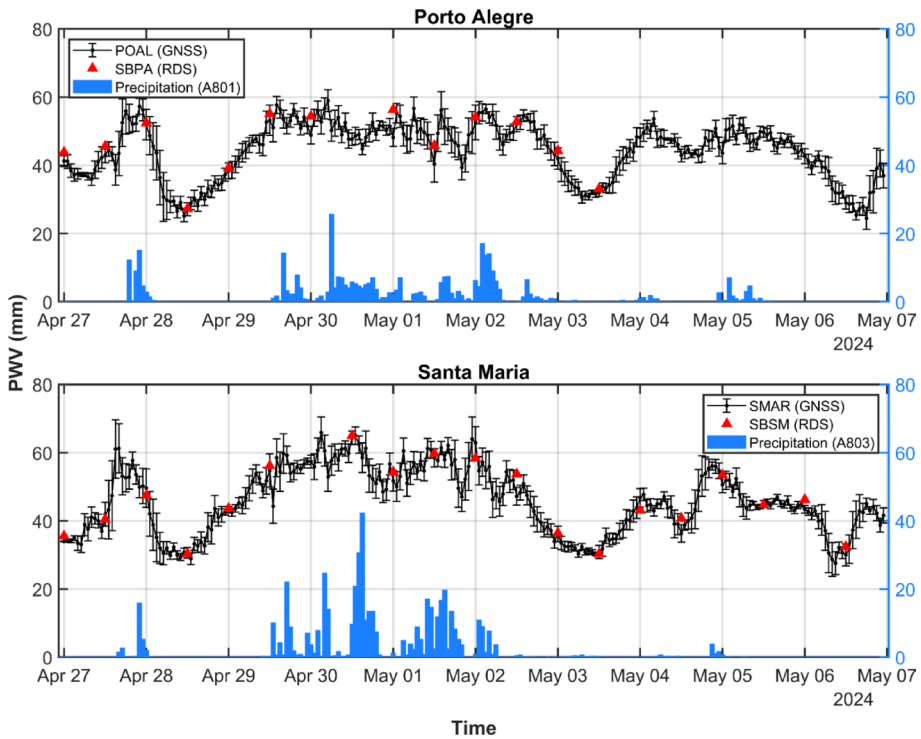


Fig. 6 PWV (GNSS, RDS) and precipitation in Porto Alegre and Santa Maria from April 27 to 7 may for 24 h (UTC). MaySource: Author (2025)

quent peak rainfall, with Santa Maria displaying higher variability (PWV standard deviation=9.52 mm; precipitation=5.37 mm) compared to Porto Alegre (PWV=8.14 mm; precipitation=3.32 mm), indicating differing patterns of atmospheric moisture dynamics. In Porto Alegre, PWV reached a maximum of approximately 60 mm on April 30, with a minimum value around 24 mm. Lower PWV values coincide with days of little to no precipitation (less than 5 mm), whereas higher PWV levels are associated with rainfall events. A similar pattern is observed in Santa Maria, where PWV peaked at around 66 mm and dropped to a minimum of 27 mm. In both locations, a strong agreement was observed between PWV estimates derived from GNSS and radiosonde data (PWV-GNSS vs. PWV-RDS), with correlation coefficients exceeding 97%. The temporal relationship between PWV-GNSS and precipitation was also evaluated. In Santa Maria, a significant correlation ($R^2=0.73$) was observed in Santa Maria, where PWV values consistently increased prior to rainfall peaks and declined afterward. This behavior confirms the potential of GNSS-derived PWV as a reliable short-term indicator of precipitation events.

Figure 7 shows the hourly values of PWV/GNSS, PWV/RDS, and precipitation for the day with the highest precipitation during the study period in Porto Alegre and Santa Maria. It also displays the error bars, where larger bars indicate periods of rapid variability, while smaller bars represent periods of greater stability.

According to Sapucci et al. (2019), peaks in PWV-GNSS values—referred to as PWV-GNSS jumps—have been observed to precede intense precipitation events within a time

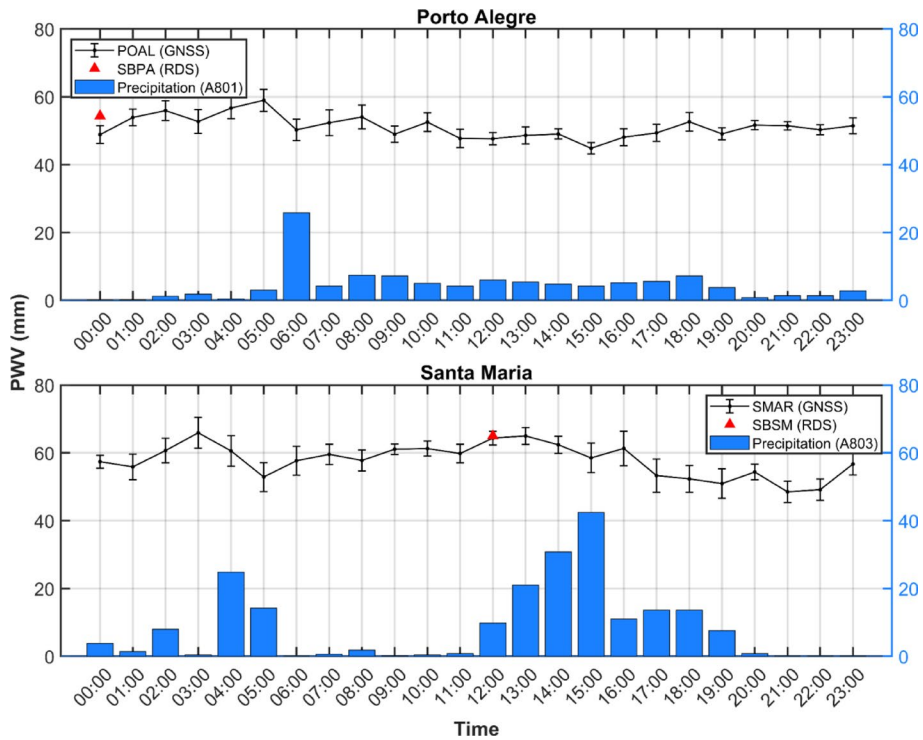


Fig. 7 PWV (PWV/GNSS and PWV/RDS) and precipitation in Porto Alegre and Santa Maria for 24 h (UTC) of the day with the highest rainfall (April 30th). MaySource: Author (2024)

window of approximately 32 to 64 min. These abrupt increases in precipitable water vapor may serve as indicators of imminent severe rainfall, highlighting their potential as a valuable tool for nowcasting applications. In this study, similar patterns were identified, with sharp increases in PWV-GNSS measurements occurring shortly before the most intense rainfall episodes, particularly in the cities of Porto Alegre and Santa Maria. Campos et al. (2023) assessed the sensitivity of GPS-PWV jumps to precipitation associated with tropical organized convective systems in Belém, northern Brazil. Their results showed a probability of detection of 84% and a false alarm rate of 25% for rainfall events exceeding 2 mm h^{-1} , confirming the potential of PWV-GNSS jumps as reliable indicators for short-term rainfall nowcasting and supporting the patterns observed in this study.

In Fig. 7, for Porto Alegre (a), a notable increase in PWV-GNSS is observed at 5:00 a.m., approximately 60 min before the onset of a 25.8 mm precipitation event at 6:00 a.m., characterizing a PWV-GNSS jump. During the precipitation period, the PWV-GNSS value drops by approximately 9 mm compared to the previous measurement, indicating the release of water vapor through rainfall. In Santa Maria (Fig. 7b), the most significant PWV-GNSS jump is recorded at 3:00 a.m., immediately preceding a sharp increase in precipitation exceeding 25 mm. It is also noted that when precipitation increases gradually (e.g., between 12:00 p.m. and 02:00 p.m.), the PWV-GNSS rise is smoother. However, the accumulated PWV-GNSS over the preceding three hours (approximately 180 mm) effectively anticipates

the intense rainfall recorded in the subsequent four hours (from 12:00 p.m. to 3:00 p.m.), with over 100 mm of accumulated precipitation.

On April 30 in Porto Alegre, PWV-GNSS and PWV-RDS measurements were available only at 00 UTC. Precipitation recorded by the automatic weather station was 0 mm, while PWV-RDS and PWV-GNSS values were 54.37 mm and 48.86 mm, respectively.

Beyond conventional meteorological applications, radiosonde data are essential for evaluating GNSS-derived atmospheric parameters. In particular, they are employed in the validation of Neutrospheric delay and PWV estimates. As GNSS signals are refracted while traversing the atmosphere—mainly due to variations in temperature, pressure, and humidity—accurate vertical profiles provided by radiosondes contribute to refining Neutrospheric delay models, enhancing the precision of GNSS-based atmospheric monitoring and weather forecasting (Bevis et al. 1992; Bock et al. 2005).

GNSS-derived PWV can be directly incorporated or assimilated into Numerical Weather Prediction (NWP) models, substantially improving the representation of initial tropospheric moisture, a key factor limiting short-term predictability. Compared to conventional nowcasting tools such as radar and satellite infrared imagery, GNSS provides continuous, ground-based, all-weather observations with high temporal resolution, complementing the spatial coverage of remote sensing systems. In operational frameworks, PWV data are assimilated using schemes such as three-dimensional variational (3DVAR) and four-dimensional variational (4DVAR) assimilation, which generate three-dimensional water vapor fields and adjust model initial conditions toward real observations (Torcasio et al. 2023; Sorkhabi & Djamour 2024a, b).

3.3 PWV alarms analysis

In this section, we assess the capability of PWV-GNSS as a predictor of rainfall at the Porto Alegre and Santa Maria stations. The verification metrics employed include the Probability of Detection (POD), which measures the fraction of observed rainfall events that were correctly forecasted; the False Alarm Ratio (FAR), which quantifies the proportion of predicted rainfall events that did not actually occur; and the Equitable Threat Score (ETS), which evaluates the skill of the predictor while adjusting for correct forecasts due to random chance (Wilks 2011; Jolliffe and Stephenson 2012).

The analysis investigates the evolution of these metrics as functions of PWV thresholds, identifying the values that achieve the optimal trade-off between detecting rainfall events and minimizing false alarms. This methodology follows established verification frameworks for categorical weather prediction and convective nowcasting (Stel & Gaiotti 2007).

To assess the predictive capability of GNSS-derived Precipitable Water Vapor (PWV) for rainfall events, a categorical forecasting framework was applied. Rainfall occurrence was defined by a threshold of 0.1 mm, distinguishing wet from dry conditions. PWV thresholds ranging from 20 to 40 mm were tested to evaluate the sensitivity of the verification metrics across different atmospheric moisture regimes.

To refine the forecast skill, temperature (30 °C) and pressure (1010 hPa) constraints were also introduced, since elevated temperatures and pressures may suppress precipitation even under high PWV conditions. For each PWV threshold, the POD, FAR, and ETS were computed, enabling a comprehensive evaluation of forecast performance and facilitating the identification of the optimal PWV threshold for rainfall prediction.

Table 4 Porto Alegre: Forecast verification metrics (POD, FAR, and ETS) for different PWV thresholds

PWV threshold (mm)	POD	FAR	ETS
20	0.983	0.506	0.027
25	0.983	0.506	0.027
30	0.983	0.482	0.074
35	0.939	0.443	0.139
40	0.913	0.379	0.245

Table 5 Santa Maria: Forecast verification metrics (POD, FAR, and ETS) for different PWV thresholds

PWV threshold (mm)	POD	FAR	ETS
20	1	0.563	0.028
25	1	0.563	0.028
30	1	0.553	0.045
35	0.881	0.527	0.081
40	0.851	0.476	0.154

The results for Porto Alegre (Table 4) show that the POD remains consistently high (>0.91) across all tested PWV thresholds, indicating that most rainfall events are successfully captured even at higher values. Meanwhile, the FAR decreases with increasing PWV, from 0.51 at 20 mm to 0.38 at 40 mm, suggesting that adopting higher thresholds effectively reduces false alarms. The ETS, which compensates for random hits, also improves with PWV, reaching its maximum value of 0.245 at 40 mm. This indicates that a 40 mm threshold provides the best trade-off between correctly identifying rainfall events and minimizing false detections. Overall, these findings suggest that a PWV threshold of approximately 40 mm represents the optimal balance for both stations, maximizing ETS while maintaining a high detection rate and acceptable false alarm levels (Table 5).

Although the ETS values are relatively low, this outcome is expected given the exploratory nature of employing GNSS-derived PWV as a single predictor of rainfall. Such low scores can be partly attributed to the limited number of precipitation events, particularly intense ones, and to the inherent variability of local meteorological conditions, which may not be fully captured by fixed PWV, temperature, and pressure thresholds. Nevertheless, the analysis effectively demonstrates the sensitivity of detection metrics to changes in PWV thresholds, revealing consistent trends in POD and FAR that offer valuable insights into the atmospheric conditions conducive to rainfall.

These findings also point to several potential improvements. Forecast skill could be enhanced by seasonally adjusting PWV thresholds, integrating complementary atmospheric variables such as humidity or wind profiles, or adopting probabilistic and machine learning approaches to better capture nonlinear relationships. Importantly, these potential refinements do not diminish the significance of the current results; rather, they emphasize that the proposed methodology establishes a robust foundation for advancing PWV-based rainfall prediction in future research.

3.4 Remote sensing of the region affected by floods

For calculating the extent of water coverage after the event, optical images from the Sentinel-2 sensor were collected as close as possible to the flooding event of April/May 2023 (Fig. 8a, b). The images indicating the situation after the flooding (Fig. 8c, d) were acquired

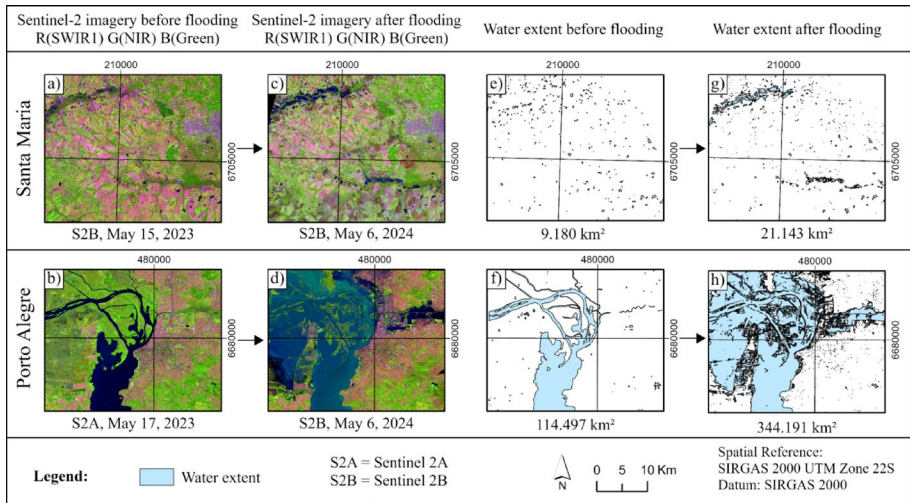


Fig. 8 Santa Maria and Porto Alegre before and after heavy rains that occurred between late April and early May 2024. **a** and **b** refer to Santa Maria and Porto Alegre, respectively, in false-color R(SWIR) G(NIR)B(Green) composition to highlight the standard water extent of the study areas. **c** and **d** show the water extent after the flooding. **e** and **f** represent the extent of water bodies mapped and provided by MapBiomass, Collection 8 (2022). Finally, **g** and **h** depict the water coverage after the flood, mapped from Sentinel-2 imagery (acquired on May 6, 2024) using a spectral water index

as close as possible to the days of heavy precipitation, May 6th, 2024, while being minimally affected by cloud cover. The figures representing the before and after of the event (Fig. 8a–d) are shown in false-color R (SWIR—1.6 μm) G (NIR—0.84 μm) B (Green—0.56 μm) Sentinel-2 composite.

For the calculation of the water-covered area, which indicates the original delineation of the water bodies in the two municipalities (Fig. 8e and f), data provided by MapBiomass Collection-8, 2022 (Mapbiomas 2024) were used.

For the determination of areas covered by water after the heavy rainfall period studied, the MNDWI (Xu 2006), was calculated using images acquired on May 6, 2024, for both cities, Santa Maria and Porto Alegre, and a threshold was applied to separate water bodies from other features.

The value resulting from the MNDWI calculation (Eq. 4) varies from -1 to 1. A threshold value of zero was further applied to extract water features from the MNDWI images. That is, the land cover type was classified as water if $\text{MNDWI} \geq 0$ and non-water if $\text{MNDWI} < 0$.

As illustrated in Fig. 8, the extent of water coverage increased significantly in the study areas following the extreme rainfall event of 2024. In the Santa Maria region, the flooded area expanded from 9.180 to 21.143 km², while in Porto Alegre, it increased from 114.497 to 344.191 km². These values underscore the magnitude of the flooding and its profound impact on the hydrological dynamics of the affected regions. Due to the lack of official ground-based flood extent mapping, validation relied on visual comparison with MapBiomass and pre-flood baselines. For comparison with GNSS data, a Sentinel-3B OLCI Level-2 LFR image covering the study area was acquired. This image, captured by the sensor on May 9, 2024, provides spatial information on Integrated Water Vapor (IWV). Figure 9 displays the spatial distribution of IWV over the municipalities of Santa Maria and Porto Alegre.

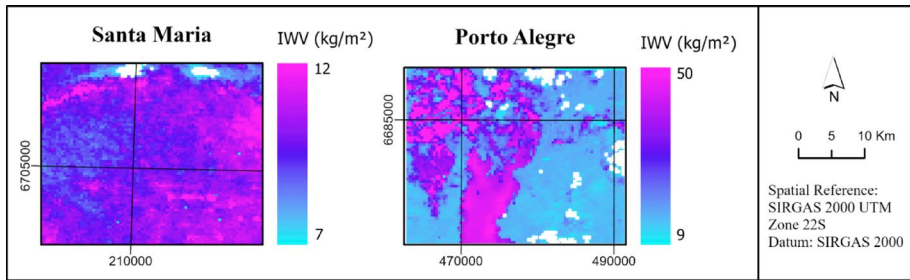


Fig. 9 Spatial distribution of Integrated Water Vapor (IWV) from the OLCI Level-2 Land Full Resolution product, obtained from the Sentinel-3 sensor on May 9, 2024, between 12:51 and 12:54 UTC, during a period of intense rainfall in the municipalities of Porto Alegre (a) and Santa Maria (b), state of Rio Grande do Sul, Brazil

As seen in Fig. 9, IWV values in Porto Alegre range from 9 to 50 kg/m², with higher concentrations observed in localized regions, indicating significant moisture availability likely associated with the occurrence of precipitation. In contrast, Santa Maria shows IWV values ranging from 7 to 12 kg/m², with a more homogeneous spatial distribution and lower moisture content compared to Porto Alegre. The average IWV estimated from Sentinel-3 for Porto Alegre was 27.75 kg/m², while in Santa Maria it was 10.25 kg/m². Although Santa Maria recorded a considerable volume of precipitation, the lower IWV values may be explained by antecedent conditions such as prior soil saturation or localized convection that released moisture before the satellite overpass. Additionally, it is important to note that Sentinel-3 has a temporal resolution of approximately two days, which may result in the underestimation of peak moisture conditions if the overpass does not coincide with maximum atmospheric water vapor availability. These limitations were considered in the interpretation of the results.

4 Final considerations

In this study, the objective was to understand the extreme event that occurred in the southern region of Brazil, specifically in the state of Rio Grande do Sul, in the aftermath of the disaster.

The tragedy was triggered by the interaction of multiple climatological systems. It resulted from a combination of climatic and synoptic-scale factors. The region was under the influence of El Niño, a phenomenon that typically increases precipitation and storm frequency in southern Brazil. Additionally, El Niño conditions intensified the Low-Level Jet (LLJ), which transports moisture from the Amazon Basin and contributes to the formation of mesoscale convective systems.

Additionally, a cold front over Rio Grande do Sul was prevented from advancing northward due to an atmospheric blocking system over central Brazil. This synoptic configuration contributed to accumulated precipitation levels reaching 303% above the climatological average in Porto Alegre and 501% in Santa Maria. Such unprecedented rainfall led to a substantial increase in soil moisture, as reported by Silveira et al. (2024), which limited the infiltration capacity of the soil and, consequently, intensified flooding in the region.

To investigate this event, we conducted a comprehensive analysis focused on the period from April 1 to May 6, 2024. Using both direct and indirect measurements, we examined the most affected areas of Rio Grande do Sul, particularly the cities of Porto Alegre and Santa Maria, mapping precipitable water vapor, precipitation, and flood extent.

The recorded disasters were associated with accumulated rainfall exceeding 457 mm in Porto Alegre and 719.6 mm in Santa Maria between April 1 and May 6, as measured by 45 automatic pluviometers. These values represent a stark contrast to those recorded during previous El Niño periods in 2015 and 2016, which reached only 150.4 mm and 143.6 mm, respectively. This comparison underscores the exceptional nature of the 2024 event and highlights the anomalous magnitude of precipitation observed in the region. On April 30, 2024, in Porto Alegre (POAL), the highest PWV value was recorded at 05:00 UTC, reaching 60 mm. At that same hour, the pluviometer at SBPA station measured 3 mm of precipitation. In the following hour (06:00 UTC), PWV sharply dropped to 10 mm, while precipitation increased significantly to 26 mm—representing a ninefold increase. In Santa Maria, throughout April 30, PWV-GNSS values remained high, surpassing 58 mm and peaking at 66 mm at 03:00 UTC, with precipitation forecasts exceeding 20 mm. At 12:00 UTC, when radiosonde data became available, the PWV-GNSS measurement was 64.35 mm, only 0.65 mm lower than the PWV derived from the radiosonde (65.01 mm), with recorded precipitation of 9 mm.

These results demonstrate that GNSS-derived PWV is a valuable tool for detecting short-term hydrometeorological risks. The clear link between PWV and rainfall supports its integration into real-time flood early warning systems in Southern Brazil. Furthermore, this work contributes to disaster risk reduction efforts and aligns with Target G of the Sendai Framework for Disaster Risk Reduction (UNDRR and WMO 2022), which emphasizes the need to substantially increase the availability of and access to multi-hazard early warning systems. The findings also support the goals of the 2030 Agenda, particularly Sustainable Development Goal 13 (SDG13- Climate Action), by promoting the use of geospatial technologies in climate adaptation strategies.

Water vapor measurements are inherently challenging to model, and these difficulties are amplified during extreme weather events. Throughout this investigation, several observations were missing due to operational limitations. To overcome these gaps, multiple data sources and methodologies were employed, allowing for broader and more accurate coverage. Each data source presents specific advantages and limitations in terms of temporal and spatial resolution, which were considered when interpreting and integrating the measurements.

PWV/RDS allowed for the evaluation of climatological variable variations across different atmospheric layers; however, the measurements were limited to only two soundings per day (when available), restricting temporal resolution. In contrast, IWV data derived from Sentinel-3 imagery provided spatially detailed observations at a resolution of 300 m, capturing atmospheric conditions at a specific moment, with a revisit time of approximately three minutes for each satellite pass. PWV/GNSS, although representing the total columnar water vapor above a station, emerges as a promising alternative due to its high temporal resolution—offering official hourly measurements and even sub-hourly sampling. Moreover, it ensures continuous data availability (24 h/day) and can be applied in nowcasting, with the ability to detect significant variations in PWV up to 60 min or less before the occurrence of precipitation events.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Tayná Aparecida Ferreira Gouveia, Helena Barbieri Azevedo, Afonso Marques Albuquerque, Maria Júlia de Souza Pompei, Aline Barrocá Marra, Viviane Aparecida dos Santos, Daniele Barroca Marra Alves and João Francisco Galera Monico. The first draft of the manuscript was written by Tayná Aparecida Ferreira Gouveia and all authors commented on previous versions of the manuscript. All authors read and approved of the final manuscript. Supervision and Corresponding author: Tayná Aparecida Ferreira Gouveia.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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